Jiarui Zhang 张家瑞

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Profile. I am a fourth-year undergraduate in the **Yao Class**, passionate about **AI infrastructure** and **ML systems**. I am dedicated to building high-performance, scalable systems for large-scale model training and inference.

EDUCATION

Tsinghua University, Beijing, China — Sep 2023 – Jun 2027 (expected)
B.Eng. in Computer Science and Technology, Institute for Interdisciplinary Information Sciences
GPA: 3.8/4.0 | TOEFL iBT: 5.5/6.0

EXPERIENCE

Student Researcher — Aug 2026 – Present
Advised by Prof. Jingzhao Zhang (Tsinghua University)

- Constructed randomized-AC^0 samplers for regular languages and Markov chains.
- Proved that revision reduces sequential computation in diffusion language models.

Research Intern — Jan 2026 – Aug 2026
Binhang Yuan Research Group, HKUST

- Developed distributed training methods for **Tree Attention**, reducing memory usage and improving throughput for multi-turn LLM training.
- Contributed to dual diffusion draft models for D^2**SD**, a speculative-decoding method for accelerating LLM inference.

Research Intern — Oct 2025 – Dec 2025
Yi Wu Research Group, Tsinghua University

- Participated in the development of a **serverless** reinforcement learning platform for scalable training.

Applied Research Intern — Jul 2025 – Sep 2025
Tencent CDG

- Designed model-based and prompt-driven reward scoring for a financial LLM, with evaluation focused on citation validity, factual accuracy, timeliness, and instruction following.
- Improved citation-format accuracy from 40% to 98%, content and timeliness accuracy from 78% to 97%, and IFEval accuracy from 38% to 80%.

PUBLICATIONS

- **Jiarui Zhang***, Yuchen Yang*, Ran Yan*, Zhiyu Mei, Liyuan Zhang, Daifeng Li, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, Binhang Yuan. "AREAL-DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models." **ICML 2026**.
- Liyuan Zhang, **Jiarui Zhang**, Jinwei Yao, Ran Yan, Yuchen Yang, Jiahao Zhang, Tongkai Yang, Yi Wu, Binhang Yuan. "D^2SD: Accelerating Speculative Decoding with Dual Diffusion Draft Models." arXiv:2606.04446.
- **Jiarui Zhang***, Chengwei Liang*, Haozhe Jiang, Binhang Yuan, Jingzhao Zhang. "Revision Provably Reduces Sequential Computation in Diffusion Language Models." Zenodo preprint, 2026.

* Equal contribution.

AWARDS

- **National Olympiad in Informatics (NOI) 2021** – Gold Medal
- **NOI 2022 Guangdong Provincial Team Selection** – 1st Place

INTERESTS

- Competitive Shuttlecock (毽球), member of the Tsinghua University Shuttlecock Team
- Hiking, especially single-day routes with ≈1,000 m of elevation gain